

Multi-Robot Formation Control System for Cooperative Warehouse Transportation

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Abstract—This paper presents the design and implementation of a multi-robot formation control system for cooperative warehouse transportation. The system assigns and coordinates small mobile robots according to the size of the target box, aiming to improve flexibility and reduce cost compared with heavier and more expensive traditional solutions. The proposed architecture integrates four main layers: perception, communication, high-level control, and low-level control. The perception subsystem uses an overhead camera, YOLO-based object detection, and ArUco markers to estimate the positions and orientations of robots and boxes in a globally calibrated workspace. The communication subsystem relies on ROS 2 and micro-ROS over Wi-Fi to transfer commands from the high-level controller to ESP32-based embedded robot controllers. At the control level, the high-level

controller combines formation control, collision avoidance, and centroid tracking, while the low-level controller performs wheel-speed regulation using encoder feedback and robust motor control strategies. The final implementation demonstrated successful two-robot formation around a target box, box carrying, and transport toward the final goal. In addition, the system achieved a lower total cost than the previous team design while improving functionality and practical robustness.

Index Terms—multi-robot systems, formation control, warehouse automation, ROS 2, micro-ROS, YOLO, ArUco, ESP32, low-level motor control

I. INTRODUCTION

Warehouse automation increasingly requires flexible robotic systems capable of handling variable payload sizes without depending on a single large and expensive robot. A cooperative multi-robot system offers a scalable alternative in which multiple small mobile robots coordinate their motion to transport boxes depending on task requirements. The main goal of this work is to develop a complete multi-robot formation control system for cooperative warehouse transportation.

The project addresses the problem of detecting a target box, assigning the required number of robots, forming the appropriate geometric configuration around the box, lifting it, and transporting it to a desired destination. A previous team reached the formation stage, where robots approached the target box and formed the required configuration, but did not complete the carrying and transportation stage. Furthermore, the previous design depended on heavier metal components, which increased structural weight and overall cost. In contrast, the current project adopted lighter metal materials, resulting in a more efficient mechanical design and a noticeable reduction in total project cost.

The complete system integrates four main layers:

- Perception layer
- Communication layer
- High-level control layer
- Low-level control layer

The main contributions of this paper are as follows:

- the design of an end-to-end multi-robot warehouse transportation system integrating perception, communication, and control;
- the development of a high-level controller combining formation control, collision avoidance, and centroid tracking;
- the implementation of a robust low-level motor controller using encoder feedback, feedforward compensation, and PID-based correction;
- the realization of a lower-cost mechanical design using lighter metal components while extending the achieved functionality.

II. SYSTEM ARCHITECTURE

The proposed architecture consists of four main subsystems: perception, communication, control, and actuation. The perception layer extracts robot and box information from the global overhead camera. The communication layer transfers processed information and control commands using ROS 2 and micro-ROS. The control layer computes the required coordinated motion commands, while the actuation layer applies the commands physically through embedded controllers, motor drivers, and wheel motors.

The robot platform includes four wheels, an ESP32 controller, DC motors with encoders, BT7960 motor drivers, and a front lifting mechanism used to engage the box after the formation is established. The modular structure allows the perception layer to operate globally, while each robot executes its own motion locally through the embedded controller.

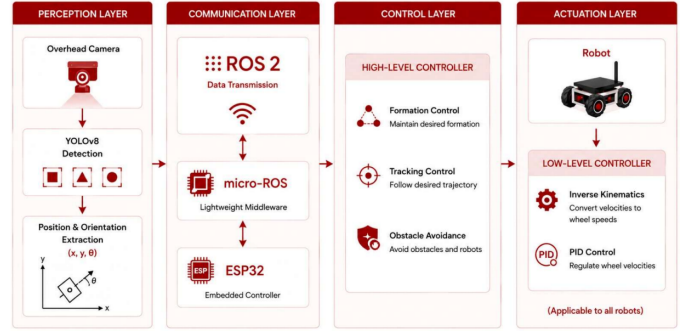


Fig. 1. Overall system architecture showing the perception, communication, control, and actuation layers.

III. PERCEPTION AND COMMUNICATION LAYERS

A. Perception Layer

The perception subsystem combines YOLO-based box detection with ArUco-based robot pose estimation. An overhead camera captures the workspace, and the robots carry mounted ArUco markers on top so that their positions and orientations can be estimated in a consistent global coordinate frame.

The selected camera operates at 720p and 30 FPS. At 30 FPS, the time per frame is approximately

$$\frac{1}{30} \approx 33 \text{ ms/frame} \quad (1)$$

which was sufficient for image processing and command generation.

The workspace was calibrated in two stages. First, intrinsic calibration corrected lens distortion using a chessboard-based method. Second, extrinsic calibration established a global workspace frame using ArUco markers and homography mapping. The final outputs included the calibration files used for undistortion and coordinate transformation.

To improve detection reliability, the dataset was prepared and augmented using Roboflow. The original dataset consisted of 200 images and was expanded to 473 images after preprocessing and augmentation. The final perception model selected for the project was YOLOv8n-OBb, since oriented bounding boxes provide better orientation-aware detection for rotated boxes.

An additional practical improvement was the use of multi-threading in the perception pipeline. Camera read time was reduced from approximately 80 ms to about 4 ms, while ArUco processing time was reduced from roughly 15–20 ms to 2–3 ms. As a result, the total perception pipeline time was reduced to nearly 15 ms per frame, which significantly improved real-time performance.

B. Communication Layer

The communication layer uses ROS 2 at the higher level and micro-ROS as the middleware bridge to the ESP32 embedded controllers. The high-level controller publishes the required commands, and each robot receives its own motion commands

over Wi-Fi using UDP communication through the micro-ROS agent.

The transmitted commands are the desired motion values:

$$(v_x, v_y, \omega) \quad (2)$$

where v_x and v_y are the commanded linear velocity components and ω is the commanded angular velocity.

This communication design allows the perception and planning computations to run on the host computer while the embedded robot controller focuses on real-time motor execution.

IV. HIGH-LEVEL CONTROL

The high-level controller computes the desired motion of each robot using the information provided by the perception layer. Based on the detected robot positions, robot orientations, and target-box location, the controller generates motion commands for each car. The controller consists of three main components: formation control, collision avoidance, and tracking control.

A. Formation Control

The formation controller is responsible for maintaining the required geometric arrangement among the robots while approaching the target box. Each robot minimizes the error between its current relative position and the desired relative displacement.

The formation-control law is

$$u_i^f = k_p \sum_{j=1}^N (p_j - p_i - \delta_{ij}) \quad (3)$$

where u_i^f is the formation-control term for robot i , k_p is the formation gain, p_i and p_j are the robot positions, and δ_{ij} is the desired relative displacement.

B. Collision Avoidance

To prevent collisions between robots, a repulsive term is activated when the robots enter a predefined safety region:

$$u_{ij}^r = \begin{cases} \alpha e^{-\beta(\|r_{ji}\| - \varepsilon r_s)}, & \text{if } \|r_{ji}\| \leq r_s \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

where α is the control gain, β is the exponential scaling factor, $r_{ji} = p_j - p_i$ is the relative position vector, and r_s is the safety distance.

The total repulsive term acting on robot i is

$$u_i^r = \sum_{j=1}^N u_{ij}^r \quad (5)$$

C. Tracking Control

Once the formation is established, the swarm centroid tracks the desired trajectory. The tracking objective is

$$\lim_{t \rightarrow \infty} (p^*(t) - \bar{p}(t)) = 0 \quad (6)$$

where $p^*(t)$ is the desired centroid trajectory and $\bar{p}(t)$ is the current centroid.

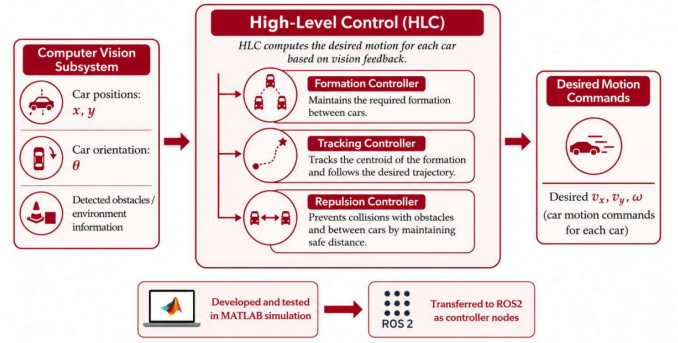


Fig. 2. High-level control visualization showing formation geometry, tracking behavior, and collision-avoidance concept.

The tracking-control law is

$$u^t = k_{tp}(p^*(t) - \bar{p}) + k_{td}(\dot{p}^*(t) - \dot{\bar{p}}) + \ddot{p}^*(t) \quad (7)$$

where k_{tp} and k_{td} are the position and velocity gains, respectively.

D. Overall High-Level Command

The complete high-level command applied to robot i is

$$u_i = u_i^f + u_i^r + u^t \quad (8)$$

Expanding the controller gives

$$u_i = k_p \sum_{j=1}^N (p_j - p_i - \delta_{ij}) + \sum_{j=1}^N u_{ij}^r + k_{tp}(p^*(t) - \bar{p}) + k_{td}(\dot{p}^*(t) - \dot{\bar{p}}) + \ddot{p}^*(t) \quad (9)$$

Before deployment on the physical robots, the high-level controller was developed and evaluated in simulation. The simulation verified that the robots could start from their initial positions, move into the required formation around the target box, track the centroid motion toward the goal, and preserve the formation near the final destination.

V. LOW-LEVEL CONTROL

The low-level controller receives target wheel-speed values and outputs PWM signals to the motor drivers. Encoder pulses are used to estimate the actual wheel RPM.

Initially, the motor-control subsystem used the L298N H-bridge. However, it suffered from dead zones, voltage drop, weak torque, and unstable low-speed response. Therefore, the system was upgraded to BT7960 motor drivers.

The final low-level strategy evolved from a basic PID-only approach into a more robust controller. The main improvements were:

- replacing the L298N driver with BT7960;
- motor-specific tuning;
- gain scheduling;
- feedforward plus PID correction;
- RPM filtering;
- target ramping;
- PWM slew limiting;
- integral clamp and anti-windup;

- retuning under real load and ground-contact conditions.

The measured RPM is computed using

$$\Delta ticks = encoderCount - lastEncoderCount \quad (10)$$

$$RPM_{raw} = \frac{\Delta ticks \times 60}{PPR \times GearRatio \times dt} \quad (11)$$

$$RPM_{filtered} = \alpha RPM_{raw} + (1 - \alpha) RPM_{previous} \quad (12)$$

The control law uses

$$error = targetRPM - RPM_{filtered} \quad (13)$$

$$FF_{low} = ff_{LowOffset} + ff_{LowSlope} \cdot targetRPM \quad (14)$$

$$FF_{high} = ff_{HighOffset} + ff_{HighSlope} \cdot targetRPM \quad (15)$$

$$integral = integral + error \cdot dt \quad (16)$$

$$derivative = \frac{error - lastError}{dt} \quad (17)$$

$$PID_{trim} = K_p \cdot error + K_i \cdot integral + K_d \cdot derivative \quad (18)$$

$$PWM_{desired} = FF + PID_{trim} \quad (19)$$

In practice, this structure provided smoother startup, reduced steady-state error, and improved low-speed performance under real operating conditions. This was especially important after adding the wheels and testing on the ground, where friction, inertia, and multi-motor operation changed the controller behavior compared with no-load bench tuning.

VI. IMPLEMENTATION AND EXPERIMENTAL VALIDATION

The final system successfully integrated perception, communication, and control into a single demonstrator. The experimental workspace covered approximately 255 cm × 180 cm, and the camera was mounted at a height of 250 cm to reduce parallax error.

The software implementation included:

- a vision node for YOLO and ArUco processing;
- custom ROS 2 messages for cars, boxes, and goals;
- a high-level control node;
- a scheduler node for sequencing the task;
- micro-ROS embedded nodes for the local robot control.

The mission flow followed the sequence:

- 1) detect robots, box, and goal;
- 2) establish the required formation;
- 3) prepare for lifting and engage the box;
- 4) track the desired motion toward the destination;
- 5) stop after reaching the goal.

The final implementation demonstrated successful two-robot formation around a target box, box lifting, and transport toward the final destination. This confirms that the current project exceeded the previous team's implementation, which had stopped at the formation stage without completing the carrying and transportation process.

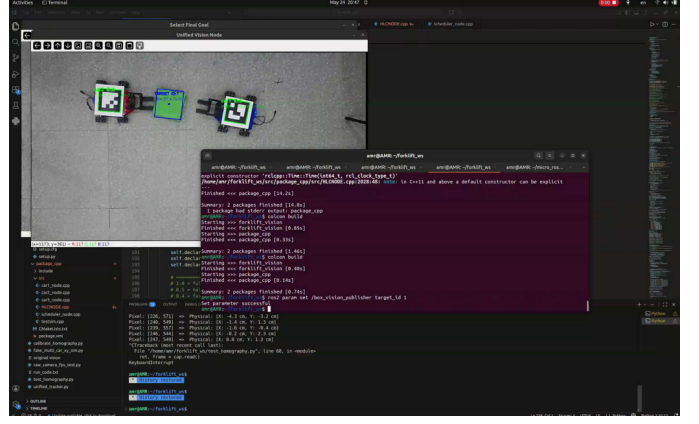


Fig. 3. Integrated system result during final operation, showing successful box handling and coordinated robot motion.

VII. DISCUSSION AND COST EVALUATION

From the perception side, the reduced pipeline time of approximately 15 ms enabled responsive vision updates. From the control side, the transition from L298N to BT7960, combined with feedforward-enhanced motor control and retuning under load, produced more stable wheel-speed regulation under real conditions. Mechanically, the use of lighter metal components improved practicality while reducing total cost.

The total project budget was 35,604 EGP, compared with 60,000 EGP for the previous team implementation. This resulted in:

- cost saving = 24,396 EGP;
- cost reduction = 40.7%.

Thus, the proposed system improved both technical functionality and economic efficiency.

VIII. LIMITATIONS AND FUTURE WORK

Although the system achieved its main objective, several practical limitations remain. The platform still depends on overhead camera visibility, and the accuracy can degrade near workspace edges because of parallax effects. Performance is also affected by changes in friction, load, and Wi-Fi communication quality. In addition, the present implementation focused primarily on the two-robot carrying case.

Future work includes obstacle avoidance expansion, further control tuning, centroid tracking for the three-car case, improved robustness against parallax, and testing under larger payloads and more cluttered environments.

IX. CONCLUSION

This paper presented a complete multi-robot formation control system for cooperative warehouse transportation. The final system combined a global overhead perception pipeline, ROS 2/micro-ROS communication, high-level formation and tracking control, and robust low-level wheel-speed regulation.

Compared with prior work, the proposed system achieved a more complete end-to-end capability by maintaining formation with two robots, carrying the box, and transporting it toward

the final goal. In addition, the system reduced the total project cost significantly through lighter mechanical design and lower-cost implementation choices.

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